MatematikRevyen 2025

von Neumann

skrevet af Baldur '20, Malthe '20

Melodi: Beyonce: "Diva"

Status: Faerdig

(3 minutter)

Roller:

I (Nadja)	Instruktør
S1 (Sirius)	Sanger
S2/Y (Baldur)	Sanger og koreograf
D1 (Anni)	Danser
D2 (Klejne)	Danser
D3 (Katrine)	Danser

(PSA starter, enten live eller som video.)

S : (stærk accent) Hello publikum, S here. We in the MatematikRevyen thought it was a shame that all the exchange students cannot follow what happens, because everything is in danish. That's why we wrote this song that EVERYONE will understand.*

*Recommended academic qualification: Academic qualifications equivalent to a BSc degree, Lebesgueintegralet og målteori, Functional Analysis, week 1-5 of Introduction to Operator algebras.

S *(optakt)*: Neumann algebra, von- von Neumann algebra
Von- von Neumann algebra, von- von Neumann algebra
Von- von Neumann algebra, von- von Neumann algebra
Von- von Neumann algebra, von- von Neumann alge-

S *(omkv)*: Von- von- von Neumann is the SO-closure of a C^*
Of a C^*, of a- of a C^*
Von- von- von Neumann is the SO-closure of a C^*
Of a C^*, of a- of a C^*

S *(vers)*: Let's refresh, so you don't stress
We get a Hilbert representation from GNS[1]

[1] Zhu, Theorem 14.4

So a C^*-algebra can be thought of as operators
Acting boundedly upon a Hilbert space full of vectors
But a function space is neat, gives rise to topology
The coarsest that gives evaluation continuity
The strong operator topology, call it SO[2]
Now we consider what we obtain by taking the closure[3]

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S *(vers)*: Got definitions, now for propositions
These algebras have polar de-compositions[4]
Got projections, whole lotta projections[5]
And you get range projections[6] with self-adjoint net convergence[7]
Give me subspace (give me subspace)
Commutating (commutating)
With all elements in $\mathcal{A}$, closed under adjoint (under adjoint)
Commutant or (or)
$\mathcal{A}'$ if you prefer[8] (If you prefer)
It's a von (It's a what?)
Neumann Algebra[9]

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S *(vers)*: Since we proved the continuous functional calculus[10]
You have asked whether we can generalize this result

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[2] Zhu, Definition 16.1
[3] Zhu, Definition 16.2
[4] Zhu, Theorem 18.9
[5] Zhu, Chapter 17
[6] Zhu, Corollary 17.6
[7] Zhu, Theorem 17.1
[8] Zhu, Definition 18.1
[9] Zhu, Proposition 18.1
[10] Zhu, Theorem 10.2

Here's the moral, there's no quarrel, for Borel we show with ease
Using a representation theorem, that of Riesz[11]
We get a measure[12], measure, now integrate $f(z)$
Over this measure, measure, and you may ask, what you get
A C^*-hom. going from, the Borel on the spectrum[13]
Of a normal element, Q.E.D. and we're done.

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S *(outro)*: Neumann algebra, von- von Neumann algebra
Von- von Neumann algebra, von- von Neumann algebra
Von- von Neumann algebra, von- von Neumann algebra
Von- von Neumann algebra, von- von Neumann algebra

[11] Zhu, Theorem 1.3
[12] Zhu, Theorem 20.2
[13] Zhu, Proposition 20.1